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# Modeling and simulation of the material removal process in electrolyte jet machining of mass transfer in convection and electric migration

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**Abstract**

Electrochemical jet machining (Jet-ECM) is a non-contact machining process which employed jetted column electrolyte as tool to impinge on the workpiece surface. Throughout the developments of this technique, the research on removal mechanism about mass transfer in Jet-ECM is of few interests. This paper presents a multi-physical approach for modeling the material removal process by coupling electrochemical reactions, fluid dynamics and mass transfer as a consequence of diffusion, convection and migration of ions in electrolyte and electrolyte-electrode interface. The electrolyte flow field is calculated for incompressible flow by Navier-Stokes equations. The distribution of ions are found to be a significant factor to current density distribution and geometry change. The consideration of mass transfer is closer to reality.

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*Keywords:* , Electrolyte jet machining, Multiphysics, Fluid dynamics, mass transfer, Current density distribution

## 1. Introduction

In the last few decades, electrochemical micromachining (ECM) is widely employed to create micro structures. ECM as a non-contact, no tool wear, and athermal progress, thus free of mechanical and thermal residue stress therefore won't initiate material property change[1]. Material removal rate could be adjusted by controlling the current applied. A wide range of metal materials, regardless hardness and thermal resistance, can be easily machined by ECM. A number of techniques based on ECM are viable to produce micro strucures, such as jet electrochemical machining.

Jet electrochemical machining(jet-ECM) employs electrolyte jet, a stream of electrolyte flows through nozzle and impinges directly on the workpiece, to localized the electrochemical reaction. Jet-ECM has demonstrated the capacity to machine various materials down to the micrometer range in [3,4], even the capability of microstructuring carbide metals [6].

Jet-ECM is a promising method to generate surface textures but suffers from relatively slow speed because of the sequential processing. Costa [2] employed textured tools consisting of arraying holes to fabricate micro dimples by Jet-ECM, presented a more efficient parallel machining process .

There are many studies on simulation of material removal process in Jet-ECM, but most of them are based on potential thus the current **i** was calculated by Ohm's law ($\mathbf{i} = \sigma\phi$), which cannot match the experimental data accurately. It is difficult to simulate the "islands"[5] in actual machining.

In the present study a model was developed with COMSOL Multiphysics to simulate the transportation of ions in convection in electrolyte flow field and the migration in electric field, i.e. the electrodynamics for a more realistic model.

## 2. Modeling of proposed method with consideration of and materials

### 2.1. Physical model

The model consists of couplings of electrochemical reactions, ion transportation, fluid dynamics and deformed geometry. Fig. 1 shows the 2D axially symmetric model. The system of equations to be solved consists of the charge conservation8, conservation for incompressible flow and ion transport.

#### 2.1.1. Electrolyte fluid dynamics

The electrolyte flow field is found to be the most influential factor in ECM process, ion transport during the electrochemical process is greatly affected by the electrolyte flow field. It is therefore of utmost importance to have an accurate flow field calculation. The electrolyte flow is governed by the lami-

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